

Turning waste wood into biodegradable circuit boards

Researchers at Empa, Switzerland's Federal Laboratories for Materials Science and Technology, have developed

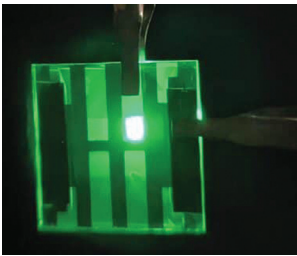


Biodegradable circuit boards (Credit: www.myscience.ch)

biodegradable printed circuit boards made from wood-based lignocellulose, offering an eco-friendly alternative to traditional fossil-fuel epoxy boards. Using natural cellulose fibres and lignin, the team creates high-performance substrates that can compost after use. Prototypes like computer mice and RFID cards function normally, while components remain recyclable. Future efforts aim to boost water resistance and scale industrial production, advancing sustainable electronics.

Paper-thin LEDs that mimic sunlight

Researchers from the American Chemical Society have unveiled paper-thin quantum dot LEDs (QLEDs) that deliver natural, sun-like indoor lighting while minimis-



A paper-thin device uses quantum dots, similar to those described in this work, to light up LEDs (Credit: Lin Zhou, Xianghua Wang)

ing sleep disruption. Unlike conventional thin LEDs that struggle to replicate sunlight, particularly yellow and green tones, these QLEDs use red, yellow-green, and blue quantum dots with zinc-sulphur shells to achieve a spectrum closely matching the sun.

Just tens of nanometres thick, the devices provide warm, eye-friendly illumination, high colour accuracy, and energy-efficient brightness for next-generation displays, adaptive lighting, and specialised applications.

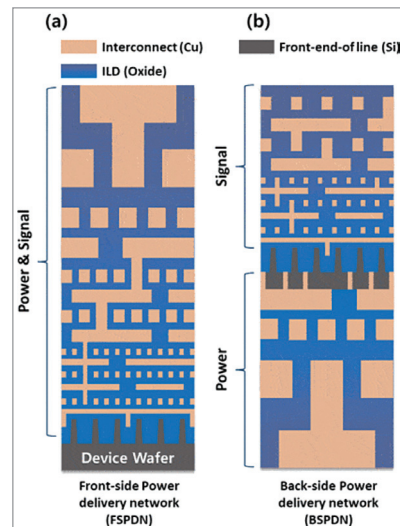
Micro-gears smaller than human hair width

Researchers at the University of Gothenburg (Sweden) have created micro-gears smaller than the width of a human hair that run entirely on laser light. These gears harness the in-

teraction between light and silicon surfaces, using deflected light to generate rotational force via conservation of linear optical momentum. Built on chips through layered silicon, patterned with light, and shaped chemically, the gears spin under laser illumination, with speed depending on light intensity. Stable, contact-free, and scalable, these 16-20µm gears promise complex chip-scale photonic systems.

Method to wire next-gen chips

As chips shrink and speed up, traditional front-side power delivery struggles with space limits and energy loss. En-



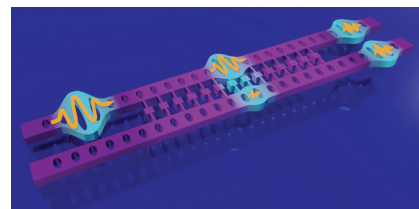
Front and back sides of the wafer (Credit: <https://ieeexplore.ieee.org>)

gineers are now turning to backside power delivery networks, routing electricity through the wafer's back to reach transistors while freeing the front for direct signal routing. Korean researchers used simulations to study wafer bending during manufacturing, finding that high heat and copper density increase stress. Balanced copper layers and

controlled heating reduce warpage, enabling thinner, reliable next-gen chips.

Chip for scalable quantum communication and sensing

Scientists at Delft University of Technology (Netherlands) have created a silicon chip that can split and control phonons, tiny quantum vibrations that carry informa-



Researchers developed a chip-based single-phonon directional coupler that carries out controllable splitting and achieves quantum-level performance (Credit: Amirparsa Zivari, Delft University of Technology)

tion like sound waves. The new four-port phononic directional coupler acts as a vibration junction, guiding energy through nanoscale patterns much like optical fibres guide light. Op-

erating at ultra-cold temperatures, it reliably splits single phonons, enabling hybrid quantum systems that link fast superconducting qubits with stable spin-based devices for scalable quantum communication and sensing.